

Board to Be Wild

Engineering Design Report

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1 Introduction

For this project a custom charcuterie board will be created. Using a wood saw, five 2 by 12 inch wood strips will be cut. The strips will then be glued together using wood glue and clamps. Using Fusion360, a 3D design will be created to engrave and mill the board. Once the design is complete, it will be deep engraved and or milled. Finally, the design will be filled with resin and set to dry.

2 Initial Design and Planning

For this project, a simple and aesthetic design would be favored. To keep the board clean, only a small thumb hole will be milled out in the bottom right corner. Then a tree design will be deep engraved. The tree will be small and starting from the side of the board. It will fit in the upper left corner and extend to the upper right corner. After deep engraving the tree, it will be filled with black resin.

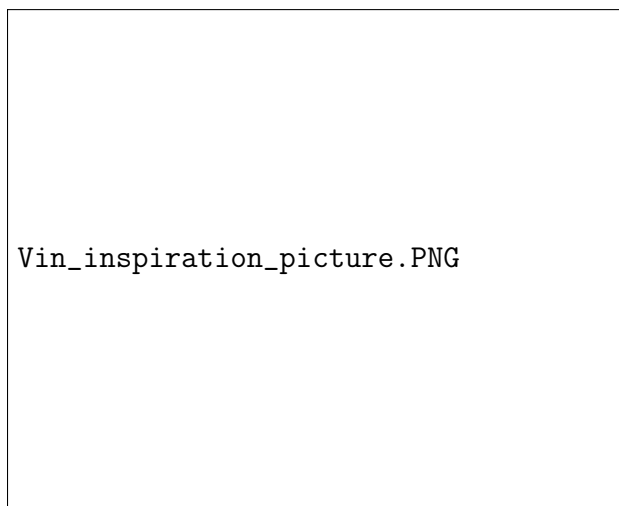


Figure 1: Inspiration picture

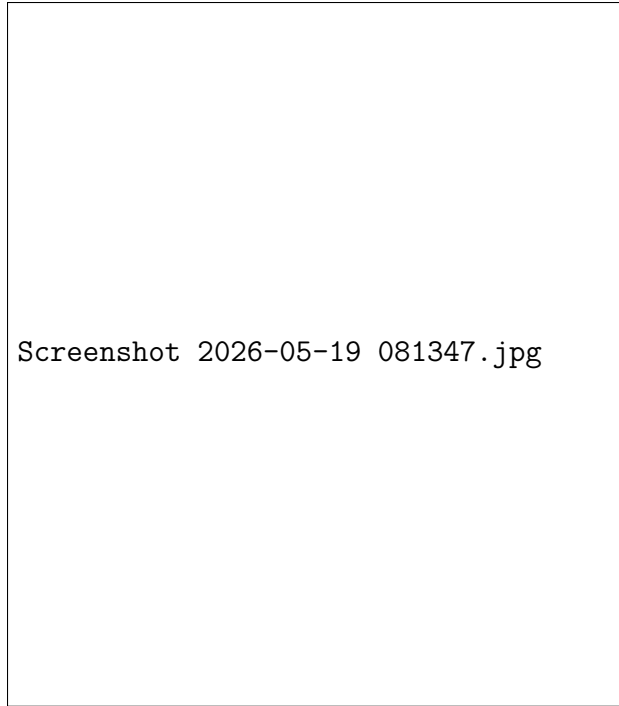


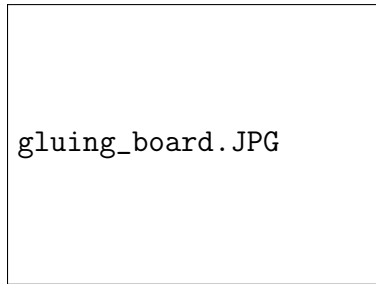
Figure 2: Original board - dimensions 10x12

3 Material Preparation

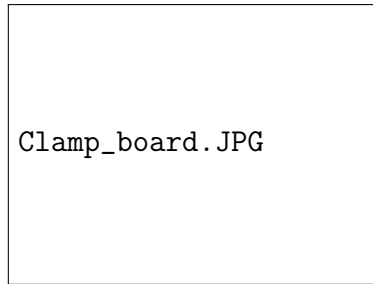
To begin the preparation process, the wood needed to be cut into 5 12x2 inch strips. Since the board will contain two types of wood, 3 strips of light wood and two of darker wood were cut using a table saw. Then, using wood glue, the strips were glued together. A paint brush was used to spread the glue evenly across the board. Then after all the strips were glued together, clamps were used to hold them tightly while drying. After letting the glue dry for a day, the wood was then planed. The planer ensured that the top and bottom surfaces were even. In addition, the planer removed any wood glue that was visible on the surface. The strips were slightly uneven in length, so the ends were trimmed with a table saw. Finally, the board was complete and ready to be milled and engraved.

4 Digital Design

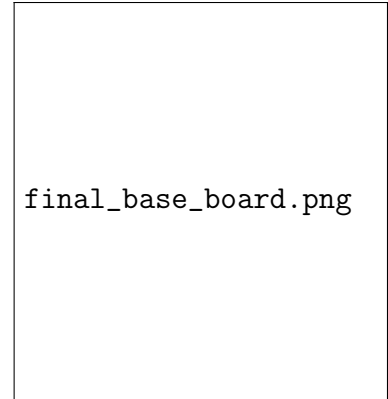
To start creating the design and making a tool path, Fusion360 was used. First, a wood board was created using the dimensions of the previously cut wood. The base board was designed to look exactly like the real one. So, the board had 5 strips of wood with alternating colors. Then a canvas of the tree design was imported onto the board. The design was positioned in the upper left corner and then negatively extruded by a quarter inch to replicate the deep engraved design. Finally, a circle sketch was created on the bottom right corner opposite from the tree and negatively extruded all the way through the board. After completing the board with the engraved design and thumb hole, the Fusion360 file was exported as a



(a) gluing board



(b) clamping board to dry



(c) Final base board - Updated dimensions 9.85x11.85

Figure 3: Material prep

DFX file. This file was then imported into Aspire to create a tool path. When the file was opened into Aspire, the physical lines of the striped board also came in. Those lines had to be deleted. In addition, it was realized that the tree design needed to also be removed because only the thumb hole was being milled. Then, a profile tool path was created for the thumb hole. The specific dimensions of the wood board that was created was inserted into the tool info. In addition, the offset and origin were selected. The offset was very specific because a jig was created to hold the board in place while milling. Finally, the tool path was completed and exported as a .nc file.

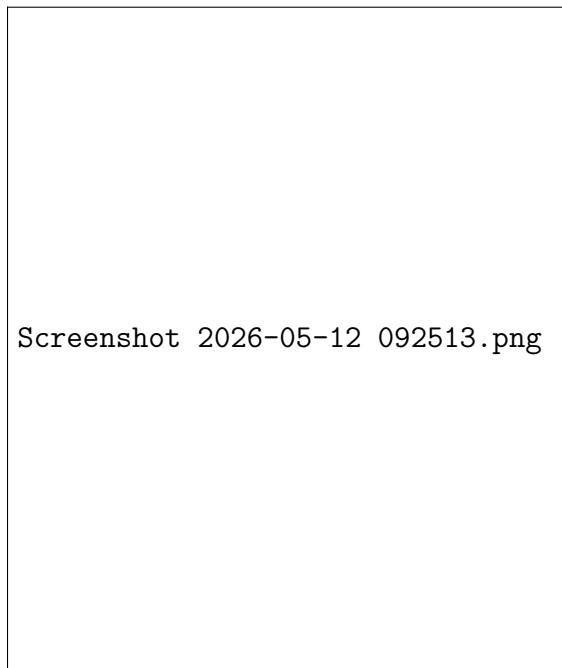


Figure 4: Final Fusion360 design

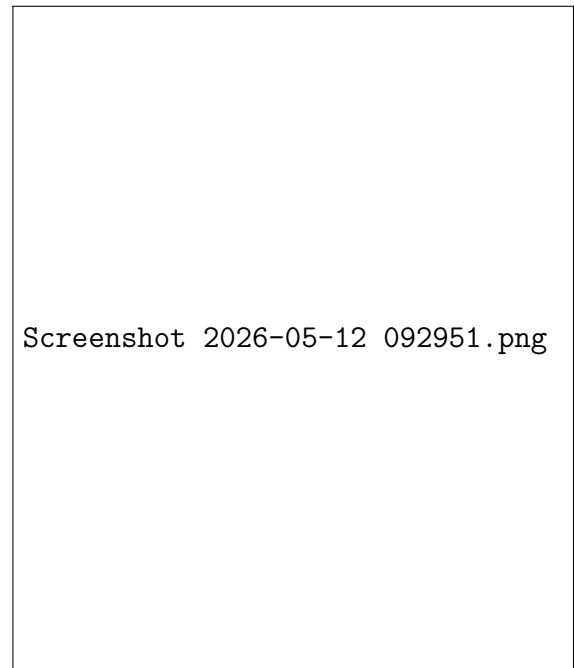


Figure 5: Imported DFX file

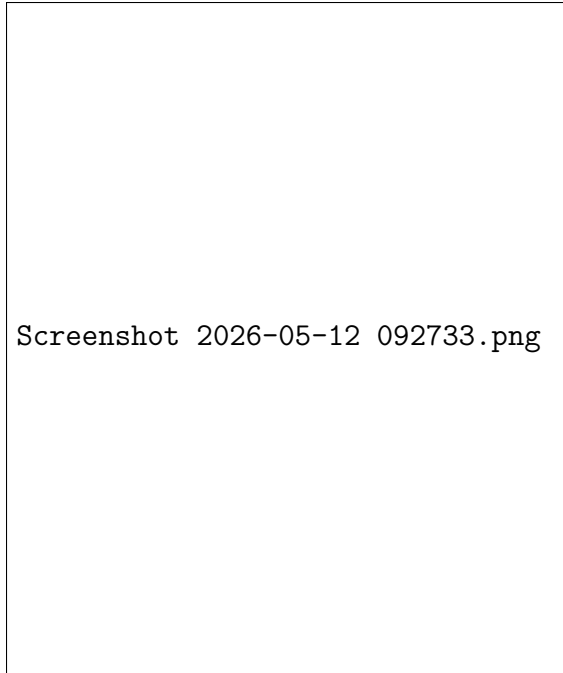


Figure 6: Cleaned up file

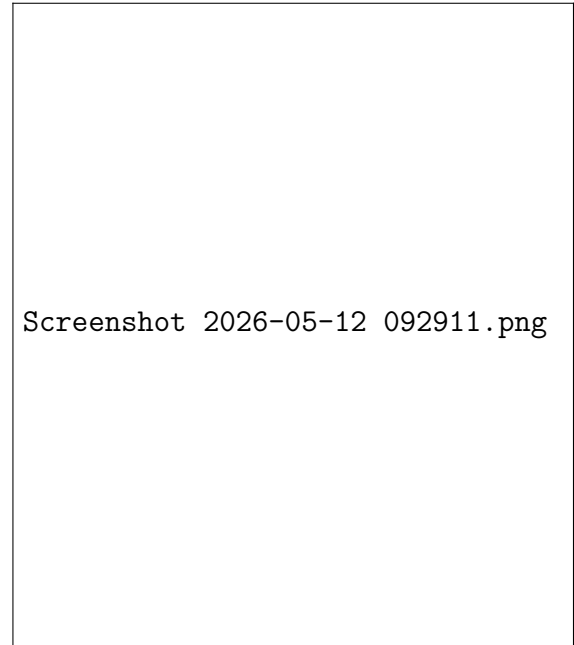
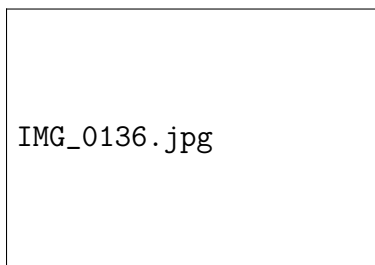


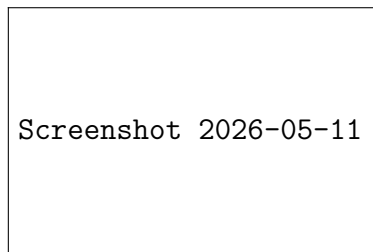
Figure 7: Final tool path

5 CNC and Manufacturing

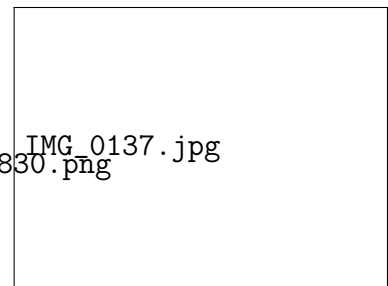
To start the manufacturing process, the Aspire tool path was uploaded on the computer with the machine. The milling machine was turned on and warmed up. Then, an air cut was run to make sure the file was correct and nothing would go wrong. After a successful air run, the machine was then leveled down to the wood board and the cut was started. It only took 30 seconds to mill out the thumb hole. Then the board was placed in the laser cutter. The tree design file was uploaded to the laser cutter and the cutter was then leveled to the board. In order to get a quarter inch of depth, the laser cutter would have to do several passes. After about 12 passes time ran out and the board was only an 8th of an inch deep. Regardless of the shallow design, the board was ready and prepped for resign.



(a) milling thumb hole



(b) laser test cut



(c) Final laser cut design

Figure 8: Manufacturing

6 Resin Process

The black resin was chosen for this design in order to look like a silhouette. First, the cup in which the resin would be poured was placed on the scale and the scale was zeroed out. Then, the resin was poured in, and the thickener was added on top. Finally, black dust was added to the cup. This mixture had to be stirred for 3 minutes straight in order to create a proper balanced solution. Once the solution was ready, a pi-pet was used to carefully drop the resin into the small details of the tree, and a Popsicle stick was used to lay the resin on the big branches. More than enough resin was added to ensure that the resin was convex and not concave. This was important because after the resin sets, it will be sanded down. After letting the resin set and dry for 24 hours, it was sanded first using the 60 grit and then the 120 grit. The entire board was then covered in water and set to dry for another 24 hours. Wetting the board made the board rough again, so the board was sanded using 180 grit and 220 grit. Finally, the board was routed to smooth the edges and then oiled to bring out the color.



Figure 9: Resin and thickener

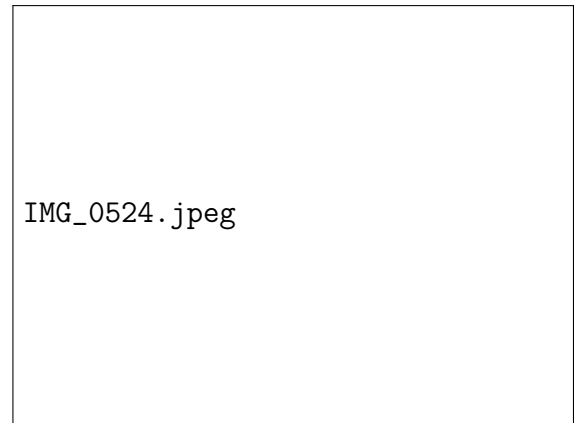


Figure 10: zero out scale with cup



Figure 11: mixing resin for 3 minutes

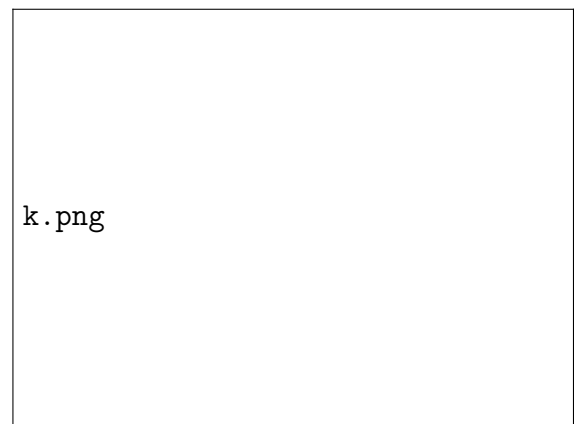


Figure 12: Final resin

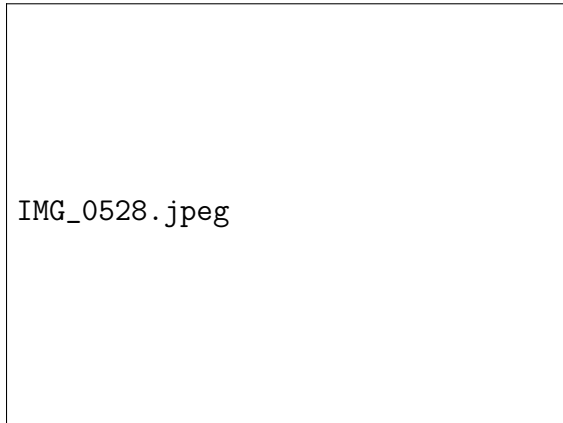


Figure 13: Sanding process



Figure 14: 60 grit, 120 grit, 180 grit, 220 grit

7 Final product

Overall, this product is of very high quality. The design is bold and the thumb whole is a good width. The edges are slightly curved, so they are not sharp and the overall board is very smooth.

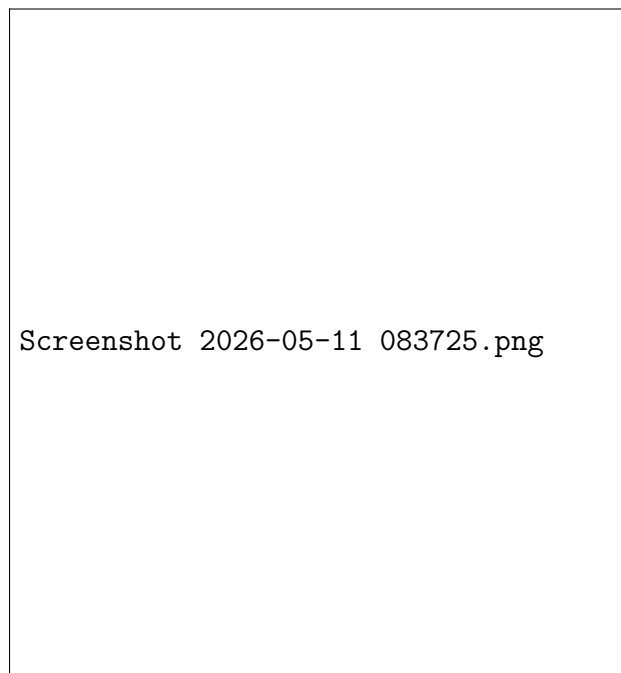


Figure 15: Final board

8 Reflection

This project tested every aspect of what was learned throughout the year. For example, one of the main units was Fusion360 design. The tools learned from this unit were applied in this project, as Fusion360 was used to map and create a 3D design of the finished product. Through this project, many challenges were faced, including designing, manufacturing, resin, and more. One specific challenge faced was deep engraving. The goal was to get the engraved design to have a depth of a quarter inch. After running the laser cutter over it many times, the design was only about an eighth of an inch deep. Under time constraints, the shallow design had to do. As a result of the design being shallow, only a thin layer of resin could be used. This ended up being a problem because after the resin was sanded down, a little bit of the wood showed through. In this project, many new tools were learned as this was the first time resin was used in a project. In addition, the table saw was used for the first time. In general, this project had all the tools learned throughout the year applied in order to make a product. Design, fabrication, manufacturing, and hands on work were all used to complete this project.